

# The Canonical V&S state

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## Abstract

Vreeswijk & Sompolinsky predict that a strongly coupled, sparse, inhibition-dominated network falls untuned into a *balanced* state: large E and I currents cancel, and the residual  $O(1)$  fluctuations drive irregular ( $CV \approx 1$ ), asynchronous, rhythmless firing, *generated* by the deterministic dynamics rather than inherited from the input. We push the conductance-based COBANet into the full four-coupling regime and test three predictions: the balanced signatures at fixed fan-in (Figure 1); their survival as  $K \rightarrow \infty$  only under strong coupling, synapse  $\propto J/\sqrt{K}$  (Figure 2); and that the irregularity is deterministic chaos, by perturbing clones on noise-free input (Figure 3). All three hold: Poisson ISIs, near-zero correlations, a broadband spectrum, a supra-Poisson CV sustained as  $K$  grows, and a positive Lyapunov exponent ( $\lambda \approx +28 \text{ s}^{-1}$  vs  $\approx 0$  for a decoupled control). The lone caveat (Figure 2) is a finite- $K$  rate drift.

## Methods

1. **Network and the  $K$ - $J$  mapping.** COBANet,  $N_E = 1024$ ,  $N_I = 256$ ,  $\Delta t = 0.25 \text{ ms}$ ,  $T = 1000 \text{ ms}$ . **Exact- $K$  initialization** ( $s = 0.99$ ): every cell begins with exactly  $K$  non-zero recurrent entries. These matrices are frozen in this experiment, so the initial pattern also remains the effective connectivity; the engine does not impose a persistent mask when such matrices are trainable. **Four recurrent matrices** use lower-clamped-Gaussian parent parameters (0.6, 0.18), (3.0, 0.9), (0.4, 0.12), and (0.4, 0.12) on the summed-coupling scale. **Per-cell independent Poisson drive** is uncorrelated (E:  $8 \text{ Hz} \times 2.10 \mu\text{S}$ , I:  $8 \text{ Hz} \times 0.25 \mu\text{S}$ ). The initialization fraction  $s$ , fan-in  $K$ , and coupling  $J$  obey

$$K = (1 - s)N_{\text{pre}} \quad (1)$$

$$J = \frac{w}{\sqrt{K}} \quad \Leftrightarrow \quad w = J\sqrt{K} \quad (2)$$

where:

- $K$ , the **fan-in**: how many presynaptic cells each neuron receives input from (the size of its “crowd”);
- $J$ , the **coupling**: the  $O(1)$  synaptic strength held *fixed* as  $K$  grows. The  $J/\sqrt{K}$  scaling is the load-bearing V&S choice, keeping the mean recurrent input  $O(\sqrt{K})$  (large, cancels) while the fluctuation stays  $O(1)$  (survives);
- $s$ , the recurrent initial-zero fraction (*-recurrent-initial-zero-fraction*);  $1 - s$  is the initially active fraction;
- $N_{\text{pre}}$ , size of the presynaptic pool a cell draws from ( $N_E$  or  $N_I$ );

- $w$ , the parent-Gaussian summed-coupling parameter passed to  $-w\text{-ei}$  and related flags, not a per-edge mean.

Exact- $K$  divides each synapse by its fan-in, so the total a cell receives,  $K \cdot (w/K) = w$ , is  $K$ -independent, giving (2). At  $s = 0.99$ :

| matrix   | direction         | $N_{\text{pre}}$ | $w$ ( $\mu\text{S}$ ) | $K$          | $J = w/\sqrt{K}$ |
|----------|-------------------|------------------|-----------------------|--------------|------------------|
| $W^{EI}$ | $E \rightarrow I$ | 1024             | 0.6                   | $\approx 10$ | $\approx 0.19$   |
| $W^{EE}$ | $E \rightarrow E$ | 1024             | 0.4                   | $\approx 10$ | $\approx 0.13$   |
| $W^{IE}$ | $I \rightarrow E$ | 256              | 3.0                   | $\approx 3$  | $\approx 1.73$   |
| $W^{II}$ | $I \rightarrow I$ | 256              | 0.4                   | $\approx 3$  | $\approx 0.23$   |

So  $K_E \approx 10$ , but the  $4\times$  smaller I pool gives only  $K_I \approx 3$ . V&S needs  $K_E, K_I$  the *same order* ( $4:1$  qualifies), but  $K_I \approx 3$  is marginal: at equal fan-in ( $K_I \approx 10$ ) the state stays balanced and asynchronous, the CV merely relaxing from 1.1–1.2 to  $\approx 1.0$ , so the supra-Poisson burstiness is a lumpy- $K_I$  shot-noise effect, not a balance failure.

2. **Why it is irregular.** Each cell sums  $\approx K$  synapses of strength  $\propto J/\sqrt{K}$ , so the mean recurrent input is  $O(\sqrt{K})$  but its fluctuation is  $O(1)$ . A moderate rate makes the large opposing E and I means **cancel to leading order**, parking the drive near threshold; the residual  $O(1)$  fluctuations carry each cell over threshold at random times (Poisson,  $\text{CV} \rightarrow 1$ ), and since each cell hears its own uncorrelated input, the population is asynchronous. These fluctuations are *network-generated*: they survive  $K \rightarrow \infty$  only under  $J/\sqrt{K}$  scaling (Step 4) and persist under noise-free input (Step 5).
3. **Diagnostics.** From post-burn-in spikes: per-neuron ISI CV, the Welch spectrum of the population-mean E trace, and the mean pairwise cross-correlogram over 100 random E pairs.
4. **The  $J/\sqrt{K}$  coupling test.** Strong coupling means synapse  $\propto J/\sqrt{K}$ , keeping an  $O(1)$  recurrent fluctuation as  $K \rightarrow \infty$ ; holding  $w$  fixed instead gives synapse  $\propto 1/K$ , the weak (mean-field) limit, fluctuation  $\propto 1/\sqrt{K} \rightarrow 0$ . At fixed  $N_E = 1024$  we sweep  $K = 10 \rightarrow 160$  under both: **strong** (weights  $\propto \sqrt{K}$ , drive folded into the balance, so rate  $\propto K$  and  $g \propto 1/\sqrt{K}$ ) and **weak** (all fixed), reading off whether the irregularity survives. (3 s trials, since the strong rate drifts low at high  $K$  and the CV needs enough ISIs.)
5. **Direct Lyapunov exponent (frozen input).** Replace the Poisson drive with a **quenched DC conductance** (a frozen per-cell offset, no per-timestep fluctuation), run two clones on *identical* input, kick every voltage by  $\varepsilon$  at  $t = 0$ , and track the **voltage distance** between the copies,

$$\|\Delta V(t)\| = \sqrt{\sum_{i=1}^{N_E} (V_i^{\text{clean}}(t) - V_i^{\text{pert}}(t))^2} \quad (3)$$

the Euclidean distance between the two copies'  $N_E$  E-cell membrane-voltage vectors ( $V_i^{\text{clean}}$  unperturbed,  $V_i^{\text{pert}}$  kicked). With noise-free input any divergence is network-generated. We compare the **balanced** four-coupling net against a **decoupled** control: the same COBA E and I cells under the same DC drive, but with all four recurrent matrices set to  $\approx 0$ , so no cell receives synaptic input from any other. Each decoupled cell is then an

independent DC-driven integrator (a clock), and a perturbation has nowhere to spread, fixing the  $\lambda \approx 0$  baseline. (It is *not* a feedforward network: there is no input-layer pathway, just the same architecture with every recurrent wire cut.) Spiking dynamics contract between spikes and expand only at spike-flips, so  $\varepsilon$  must flip spikes ( $\varepsilon = 0.1$  mV; smaller just contracts  $\rightarrow$  spurious  $\lambda < 0$ ). The slope of  $\log\|\Delta V\|$  in the post-flip, pre-saturation window is the largest Lyapunov exponent (five seeds). It is an *initial-growth* estimate ( $\|\Delta V\|$  saturates at the attractor size), so the sign, not the magnitude, is the result.

## Results

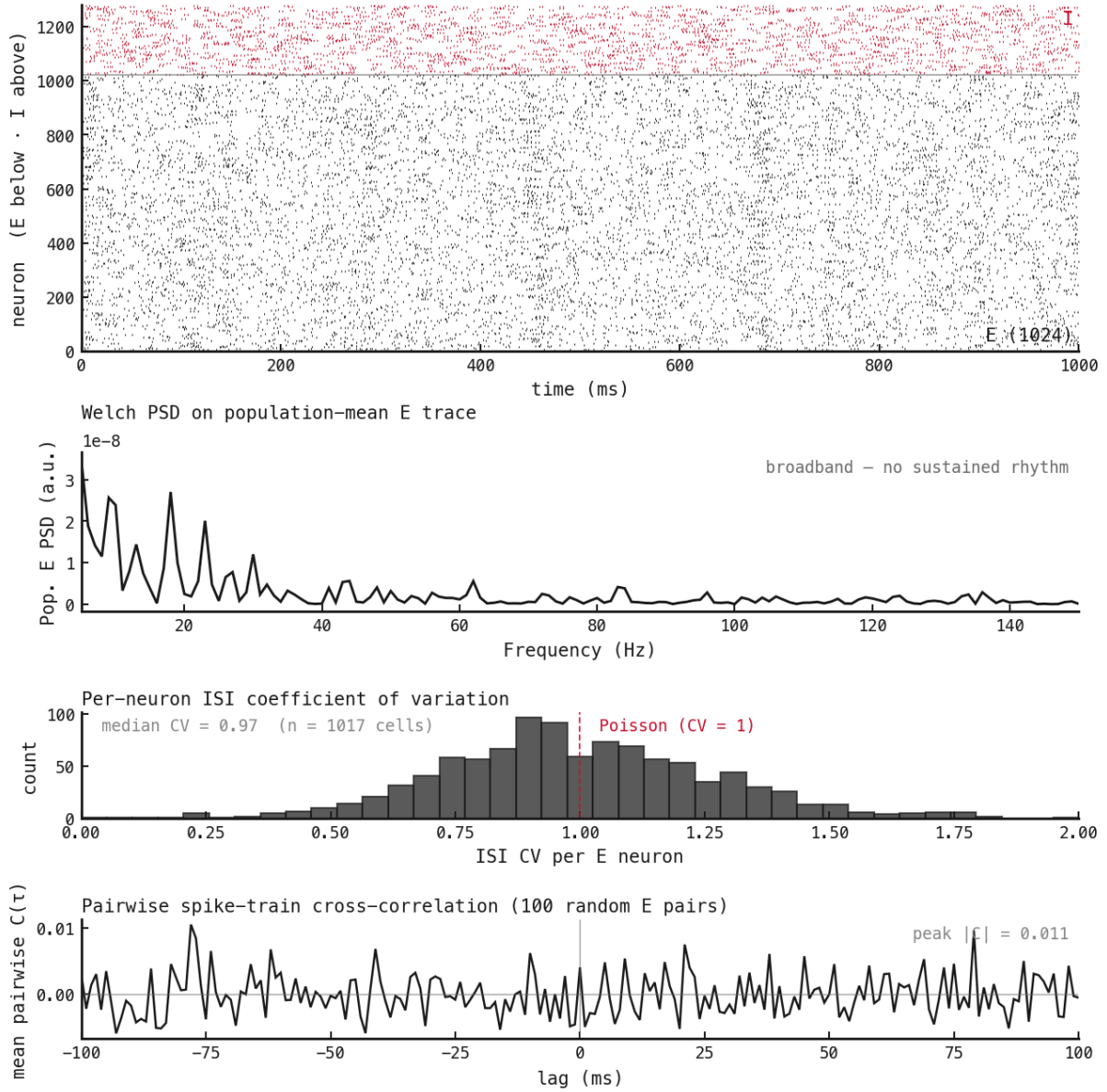


Figure 1: Four diagnostics of the four-coupling state (E black, I red above the divider;  $\langle r_E \rangle \approx 17.5$ ,  $\langle r_I \rangle \approx 25.4$  Hz). **What we expect.** Irregular (ISI CV  $\approx 1$ ; a constant-current cell sits at 0), asynchronous (near-zero pairwise correlation), rhythmless (broadband, no peak). **What we see.** *Raster*: scattered, no bands. *PSD*: broadband, no sustained rhythm; the single-trial spectral max wanders seed to seed (5–90 Hz across six seeds), so it is not a peak, but a *weak* gamma-band E–I resonance does survive seed-averaging ( $\approx 2\times$  the floor near 40 Hz, the same  $E \leftrightarrow I$  loop that sharpens into PING gamma, here heavily damped). *ISI CV*: median 1.11 (E), 1.20 (I). *Cross-correlogram*: flat, peak  $|C(\tau)| \approx 0.01$  despite heavy shared input, the active decorrelation of Renart et al. (2010). **Do they align?** Yes on all three: the COBANet enters the balanced state untuned.

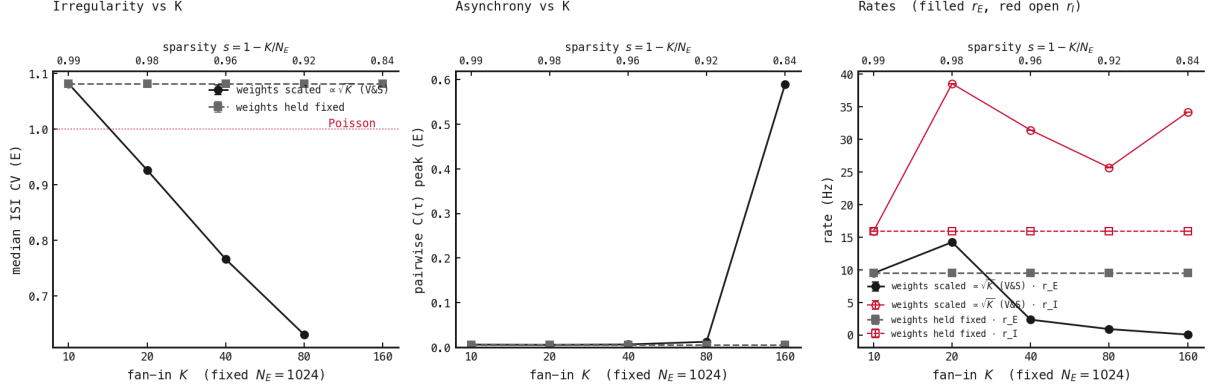


Figure 2: Fan-in  $K = 10 \rightarrow 160$  at fixed  $N_E = 1024$ , equivalently sparsity  $s = 1 - K/N_E$  from 0.99 to 0.84 (top axis), with recurrent weights scaled  $\propto \sqrt{K}$  (black, the V&S  $J/\sqrt{K}$  rule) versus held fixed (grey);  $r_I$  shown in red.  $\pm 1$  SD over three seeds, 3 s trials. **What we expect.** Strong coupling keeps the  $O(1)$  fluctuations, so irregularity *survives* as  $K$  grows; weak coupling loses them ( $\propto 1/\sqrt{K}$ ), so cells *regularise*. Asynchrony holds in both. **What we see.** *Irregularity:* strong stays supra-Poisson (CV 1.2–1.3, easing to 1.11 at  $K = 160$ ); weak decays to  $\approx 1.0$ , the Poisson floor. *Asynchrony:* both  $\approx 0.006$  throughout. *Rates:* strong  $r_E$  drifts  $17.5 \rightarrow 4.9$  Hz (the  $O(1/\sqrt{K})$  finite- $K$  correction); weak holds  $\approx 17$  Hz. **Do they align?** Yes: only  $J/\sqrt{K}$  keeps the recurrent irregularity alive as  $K$  grows, and the gap is the signature. The one mismatch is the strong-coupling rate drift (Next steps); whether the survival is *self-generated* is settled by Figure 3.

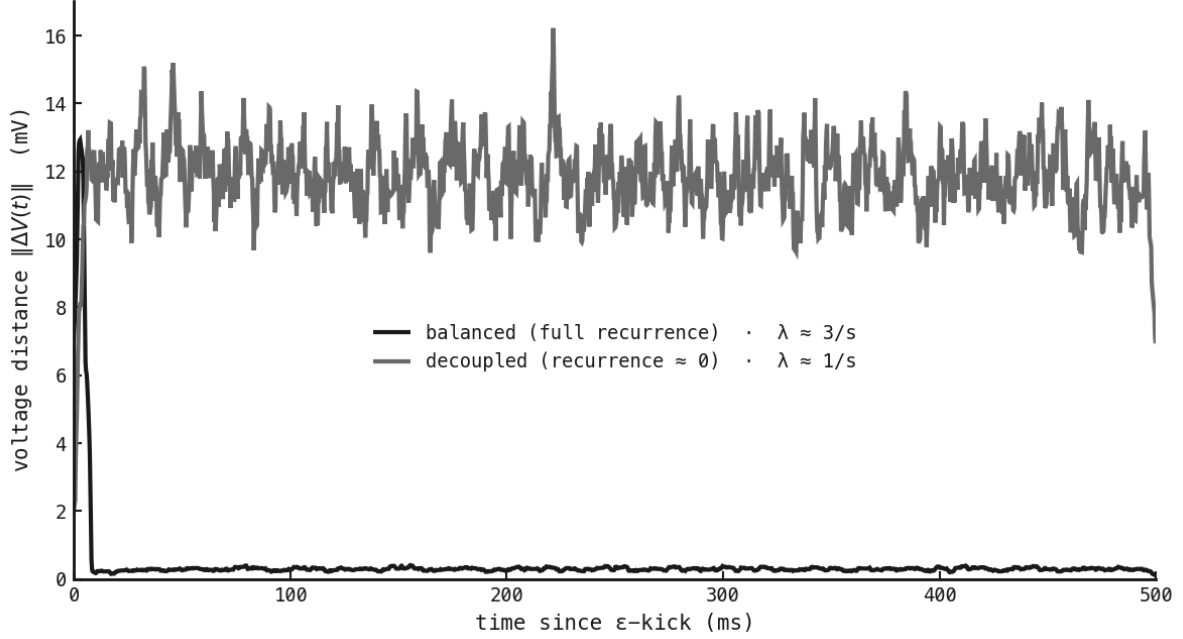


Figure 3: Two clones on *identical* quenched-DC input, the second kicked by  $\varepsilon = 0.1$  mV at  $t = 0$ ; voltage distance  $\|\Delta V(t)\|$ , seed-mean (bold) with a  $\pm 1$  SD band over five seeds, lightly smoothed. **What we expect.** If chaotic, the kick is *amplified* ( $\|\Delta V\|$  grows exponentially,  $\lambda > 0$ ), and with noise-free input that can only be network-generated. The decoupled control should not grow ( $\lambda \leq 0$ ). **What we see.** *Balanced (black)*:  $\|\Delta V\|$  rises steeply (fitted rate  $\lambda \approx +28$  s $^{-1}$ ) then saturates at the attractor size ( $\approx 250$  mV). *Decoupled (grey)*: neither amplified nor forgotten, a fixed phase offset,  $\lambda \approx 0$ . **Do they align?** Yes:  $\lambda > 0$  for the balanced net,  $\approx 0$  for the control, deterministic chaos, self-generated. Caveat: an *initial-growth* estimate (saturation caps the window), so the sign, not the magnitude, is the result; a renormalised Benettin scheme would pin exact  $\lambda$  (Next steps).

## Next steps

Figures 1–3 confirm the balanced signatures, their  $J/\sqrt{K}$ -only survival, and a positive Lyapunov exponent: the state is self-generated and deterministically chaotic. Two refinements remain.

1. **Hold the rate as  $K$  grows.** Strong-coupling  $r_E$  drifts  $17.5 \rightarrow 4.9$  Hz (the  $O(1/\sqrt{K})$  finite- $K$  correction). A DC offset to E alone *fails*: it pushes E off the balanced point and inflates CV to 1.4–1.7. Honest pinning must co-adjust the E *and* I drives along the balanced manifold (solve the 2D balance for a target  $(r_E, r_I)$ ), giving a clean constant-rate CV comparison.
2. **Pin down the asymptotic  $\lambda$ .** Figure 3 gives the sign; the magnitude is an initial-growth estimate ( $\varepsilon$ -dependent, saturation-capped). The exact exponent needs a *renormalised* Benettin scheme (lockstep clones, periodic rescaling of  $\Delta V$ , averaging log-growth over a long trajectory), which also yields the full spectrum and the attractor dimension.